PAPER_203: UQFF CMB, Structure Growth, Non-Gaussianity, and Curvature Perturbation

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Author: Star-Magic UQFF Research Framework
Source: grok_share_7514fe.txt lines 6080–6095 (BB_C_Equations items 1380–1430)

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\mathrm{day}^{-1},\ [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM_s}{r^2}, \quad \text{with } \kappa = 5.0 \times 10^{-4}\,\mathrm{day}^{-1},\ [SSq] = 0.57$$

Abstract

This paper applies the UQFF framework to inflationary and post-inflationary perturbation physics: primordial non-Gaussianity (f_NL), single-field slow-roll inflation curvature spectrum, post-inflationary reheating, structure growth factor D(a), and the LQC pre-bounce curvature perturbation modification. The unified UQFF perspective embeds all of these into F_UBii operators with d_c Gaussian tails, allowing consistent statistical comparisons across CMB, LSS, and LQC regimes.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-}4 day^{-1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Non-Gaussianity Parameter f_NL

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Local non-Gaussianity (single-field slow-roll):
  f_NL = 5/6·(G³ - 3G·G?² + 2·G?³)/G4

  where: G = field velocity G(?) in Dirac-Born-Infeld models
  Standard single-field: f_NL = (5/12)(ns - 1) ~ -0.03 (undetectable)
  Multi-field/curvaton: f_NL ~ O(1–100) (potentially observable with CMB-S4)

F_UBii,ng = F_rel × (f_NL × d_c³ / E_LEP) × Q_wave × exp(-d²_c/(2s²))

Um,ng(f) = µ(?_vac)·(1-e^{-?t})·[from d? curvature on superhorizon scales]

Planck 2018 bound: f_NL,local = -0.9 ± 5.1  (1s, no detection)
CMB-S4 forecast: s(f_NL) ~ 1–2  (improved constraint)
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2. Primordial Curvature Power Spectrum

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Single-field slow-roll inflation:
  P_R(k) = H²/(8p²e·M²_Pl)  ~ 2.1×10??   (at k0 = 0.05 Mpc?¹)
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Spectral index and tilt:
  n_s = 1 + d ln P_R/d ln k = 1 - 6e + 2?   (to first order in slow-roll)
  Planck 2018: n_s = 0.9649 ± 0.0042 (>5s detection of tilt)

Running (scale-dependent tilt):
  dn_s/d ln k = -16e? + 24e^{2} + 2?^{2} (second slow-roll order)

Tensor-to-scalar ratio:
  r = 16e   (BICEP/Keck: r < 0.036 at 95% CL, 2021)

F_UBii,curv = F_rel × (P_R(k) / E_LEP) × Q_wave × (d_c/s)

Um,curv(?) = μ(?_vac)·(1-e^{-?t})·[?? = v(2e)·H·M_Pl]

UQFF connection: vacuum energy ?c^{2}/3 modifies P_R at large scales (low multipoles)
  P_R,UQFF(k) = P_R(k)·(1 + ?UQFF·c^{2}/(3H^{2}))

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3. Reheating Evolution

End of inflation: ? oscillating around minimum, V(?) ~ (1/2)m^{2}?^{2}

Reheating temperature:
  T_reh = (30V_end/(p^{2}g_*))^{1/4} · e^{-3N_reh/4}

where:
  V_end = inflaton potential at end of inflation
  N_reh = number of e-folds of reheating
  g_* = effective DOF at reheating (~100–200 for SUSY)

Radiation domination begins when G_inf = H (inflaton decay rate equals Hubble):
  T_reh,min ~ (90/(8p^{3}g_*))^{1/4} · v(G_inf·M_Pl)

F_UBii,reh = F_rel × (T_reh / E_LEP) × Q_wave × [g_* and N_reh as free parameters]

Um,reh(N) = μ(?_vac)·(1-e^{-?t})·(30V_end/(p^{2}g_*))^{1/4}·e^{-3N_reh/4}

BBN constraint: T_reh > T_BBN ~ 4 MeV (required for successful nucleosynthesis)
Gravitino constraint: T_reh < 10? GeV (SUSY, avoid gravitino overproduction)

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4. Structure Growth Factor D(a)

Linear growth equation:
  d̈ + 2H(a)·d? = (3/2)·O_m·H^{2}(a)·d/a^{3}

Growing mode solution:
  D(a) = (5O_m/2) · H(a)/H0 · ?0^a da'/[a'H(a')/H0]^{3}

Growth rate:
  f = d ln D/d ln a ~ O_m(a)^{0.55}   (Linder 2005 approximation)

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F_UBii,grow = -F_rel × (D(a)·d0 / E_LEP) × Q_wave × f(O_m)

Um,grow(a) = µ(?_vac)·(1-e^{-?t})·[Growing mode D ? a in matter era, suppressed by DE]

Key values:
  D(z=1)/D(z=0) ~ 0.76 (matter + ? cosmology)
  s_8 = 0.811 ± 0.006  (Planck 2018)
  f·s_8 ~ 0.46 at z=0   (RSD measurements)
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5. LQC Pre-Bounce Perturbation Modification

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Standard primordial power spectrum:
  P(k) = A_s·(k/k0)^{n_s-1}

LQC pre-bounce modification (Dapor-Liegener approach):
  P_LQC(k) = P(k) · (1 + k/k_*)^{-a}

where:
  k_* = quantum bounce scale (k_* ~ k_Pl/?_bounce ~ 10?^2 Mpc?^1)
  a = UV suppression exponent (a ~ 2–4)

Physical interpretation:
  - For k << k_*: P_LQC ? P (standard CMB, no modification)
  - For k >> k_*: P_LQC ? k^{n_s-1-a} (suppressed at superhorizon/Planck scales)
  - Provides natural large-scale power suppression (low-l CMB anomaly)

F_UBii,lqcp = -F_rel × (P_LQC(k) / E_LEP) × Q_wave × (1 + k/k_*)^{-a}

Um,lqcp(k) = µ(?_vac)·(1-e^{-?t})·[Power tilt + UV suppression at Planck modes]
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6. Sakharov Oscillations and BAO

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Baryon Acoustic Oscillations (BAO) peak scale:
  r_s(z_d) = ?0^{z_d} c_s dz/H(z)

  c_s = c/v(3(1 + 3?_b/(4?_?)))    (sound speed before decoupling)
  z_d ~ 1020  (drag epoch)
  r_s ~ 147 Mpc  (physical BAO scale today)

BAO detection:
  Angular diameter distance D_A(z) = r_s·?_BAO
  Hubble D_H(z) = r_s/?z_BAO

UQFF BAO connection:
  ?_J in baryon-photon fluid sets r_s ? same Jeans mechanism as F_UBii,jeans
  But: ? = ?_b + ?_? >> ?_gas ? ?_J,BAO >> ?_J,gas
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7. CMB Polarization and Tensor Modes

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E-mode polarization from density perturbations:
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  C_l^{EE} = (2/p)?k²dk·P(k)·|?_l^E(k)|²

B-mode from primordial gravitational waves:
  C_l^{BB} = (r/16)·C_l^{tensor}    (proportional to tensor-to-scalar ratio r)

B-mode from lensing:
  C_l^{BB,lens} = ?d²l' (l'·e^)²·C_{|l-l'|}^{EE}·C_{l'}^{??}

UQFF role in polarization:
  The oscillating FU_Bi_i buoyancy at epoch of last scattering generates
  a correlation between curvature and polarization through:
  d_T/T|_Doppler = v_b·n^  (velocity perturbation from baryon motion)
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8. Summary: Perturbation Chain in UQFF

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Inflation
  ? e, ? (slow-roll)
  ? F_UBii,curv : curvature seed P_R(k)
  ? F_UBii,ng   : non-Gaussianity f_NL correction
     ? reheating
  ? F_UBii,reh  : thermal equilibration T_reh
     ? BBN
  ? F_UBii,deb + F_UBii,eta : light element abundances
     ? recombination/CMB
  ? F_UBii,cmb + F_UBii,recomb : photon decoupling
     ? structure formation
  ? F_UBii,grow : linear growth factor D(a)
     ? reionization
  ? F_UBii,ion + F_UBii,bub : HII bubble percolation
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Each stage connects through Q_wave × (F_X/E_LEP) common factor, enforcing 99.9% backbone unification across all 99 UQFF systems.

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9. Numerical Values

Parameter	Value	Source
n_s	0.9649 ± 0.0042	Planck 2018
A_s	2.1×10??	Planck 2018
r	< 0.036	BICEP/Keck 2021
f_NL,local	-0.9 ± 5.1	Planck 2018
s_8	0.811 ± 0.006	Planck 2018
r_s,BAO	147 Mpc	Eisenstein et al.
T_reh (BBN lower bound)	> 4 MeV	Standard BBN

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10. References

- `grok_share_7514fe.txt` lines 6080–6095 (BB_C_Equations items 1380–1430)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)

- PAPER_202: UQFF Reionization, BBN, Recombination
- Planck 2018 I–X papers (cosmological parameters)
- BICEP/Keck 2021 (B-mode constraints)